UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – Fall 2021

Roll No. ....

Paper: Advanced Mathematical Economics **Course Code: ECON-307-A** **Time: 3 Hrs. Marks: 60**

Q.1. Answer the following short questions: **(6x5=30)**

I. Explain the concept of Maximum Principle.

II. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$

III. Find characteristic roots of: $(r+4)(r+4)(r^2 + 2r + 2) = 0$

IV. Write complementary function of characteristic equation $(r+4)(r+4)(r^2+2r+2) = 0$.

V. Explain the concept of Primal and Dual.

VI. Explain Two Variable Phase Diagrams.

Answer the following questions. **(3x10=30)**

Q. 2. Test convergence of the path of $y_{t+2} - 8y_{t+1} - 2y_t = 10$, using Schur Theorem.

Q. 3. Using Routh Theorem, find dynamic stability of $y'''(t) + 4y''(t) + 5y'(t) - 2y = -2$

Q. 4. Write Kuhn Tucker Conditions for numerical, $C = (x_1 - 4)^2 + (x_2 - 4)^2$, $2x_1 + 3x_2 \geq 6$, $-3x_1 - 2x_2 \geq -12$, $x_1, x_2 \geq 0$. Further solution not required.